

Practice Quiz 03D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 03

Nested quantifiers and negating quantified statements.

1. Consider

$$(i) \forall x \in \mathbf{R} \exists y \in \mathbf{R} (y = x^2) \quad (ii) \exists y \in \mathbf{R} \forall x \in \mathbf{R} (y = x^2).$$

(a) State each in plain English and say whether it is true or false.

Solution Statement (i) says that the square of every real number is a real number. It is true: for a given x , take $y = x^2$. Statement (ii) says that one fixed real number equals the square of every real number. It is false. ■

(b) Negate (ii), with no negation symbol to the left of a quantifier.

Solution Its negation is

$$\forall y \in \mathbf{R} \exists x \in \mathbf{R} (y \neq x^2).$$

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